Physical ROS-Generating Modalities as Spatially Selective Ferroptosis Inducers for Drug-Tolerant Persister Cells: A Cross-Literature Analysis of 10,413 Articles

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Abstract

Drug-tolerant persister cells that survive first-line cancer therapy are sensitized to ferroptosis — an iron-dependent cell death pathway. The persister-ferroptosis field currently focuses exclusively on pharmacologic ferroptosis inducers (RSL3, erastin, GPX4 inhibitors) for clinical exploitation. Here we present a cross-literature analysis of 10,413 research articles (1,668 journals, 19 mechanisms, 22 cancer types, 2015-2026) proposing that an entire class of physical ROS-generating modalities should be evaluated as spatially selective alternatives. Photodynamic therapy (PDT, 355 PubMed ferroptosis articles) and sonodynamic therapy (SDT, 121 articles) trigger ferroptosis through sensitizer-activated ROS generation, and additionally produce immunogenic cell death (ICD) — a dual output that pharmacologic inducers have not been shown to reliably generate. PDT is established but depth-limited (light penetrates millimeters); SDT extends the same mechanism to deep-seated tumors via ultrasound (centimeters). We propose that physical modalities offer two advantages over systemic drugs for persister targeting: spatial selectivity (focused energy at the tumor, not systemic exposure) and ICD generation (potential immunotherapy synergy). This framing — which connects two research communities that do not cite each other — has not been systematically proposed. We present it as a research direction, not a validated strategy, and discuss extensively the PDT precedent (40 years without demonstrating robust ICD-immune synergy in randomized trials), the nanosonosensitizer confound, failed trials (CheckMate-498, BIND-014, Pexa-Vec), and the biological barriers that could prevent translation.

Keywords: drug-tolerant persisters, ferroptosis, photodynamic therapy, sonodynamic therapy, immunogenic cell death, physical modalities, ROS, spatially selective therapy, cross-literature analysis

1 Introduction

The past decade has witnessed a substantial diversification of cancer therapeutic mechanisms. Where oncology was historically dominated by the triad of surgery, chemotherapy, and radiation, the contemporary landscape encompasses at least 19 mechanistically distinct approaches ranging from bioelectric modulation and frequency-based physical therapies to CRISPR-mediated gene editing, personalized mRNA cancer vaccines, and microbiome manipulation [Y et al. [2022a]; ZE et al. [2022]; LA et al. [2023]]. This expansion has been accompanied by a fundamental conceptual shift: the recognition that cancer's hallmark heterogeneity, adaptive resistance, and microenvironmental complexity render single-mechanism cures unlikely for most malignancies, and that rational multimodal combinations may be required to achieve durable responses [S et al. [2021]].

Several developments have accelerated this convergence. The success of immune checkpoint inhibitors, beginning with the landmark approvals of ipilimumab, pembrolizumab, and nivolumab, established that mobilizing endogenous anti-tumor immunity could produce durable remissions even in previously intractable cancers [Y et al. [2022a]]. Yet response rates to single-agent checkpoint blockade remain modest across many tumor types, with primary and acquired resistance affecting the majority of patients [L et al. [2023]]. This has driven systematic exploration of combination strategies: checkpoint inhibitors with oncolytic viruses [SZ et al. [2023]], with CAR-T cell therapy [D et al. [2025]; L et al. [2024]], with metabolic reprogramming [C et al. [2023]], and with physical modalities such as tumor-treating fields (TTFields) [O et al. [2021]; D et al. [2022]] and irreversible electroporation [Ringel-Scaia et al. [2019]].

Simultaneously, the maturation of platform technologies – lipid nanoparticles for mRNA delivery [J et al. [2022a]], CRISPR-Cas9 for precision gene editing [A et al. [2022]; J et al. [2022b]], and antibody-drug conjugates for targeted payload delivery [F et al. [2025]] – has created a combinatorial toolkit of extraordinary breadth. The question confronting the field is no longer whether multi-mechanism approaches are desirable, but rather which specific combinations, for which cancers, with what evidence, and where the most consequential gaps lie.

Despite this imperative, to our knowledge, no systematic effort has mapped the full mechanism-cancer-evidence landscape across the breadth of therapeutic modalities now under investigation. Individual reviews have addressed specific domains – immunotherapy combinations [S et al. [2021]], nanomedicine [C et al. [2025]; H et al. [2025a]], oncolytic virotherapy [R et al. [2023]], or TTFields [O et al. [2021]] – but a cross-cutting synthesis that identifies convergence patterns, translational bottlenecks, and unexplored opportunities across the entire therapeutic spectrum has been lacking.

This article presents a systematic analysis of 10,413 cancer research articles spanning 19 therapeutic mechanisms and 22 cancer types, drawn from 1,668 journals (2015-

2026). While the Results section provides a comprehensive landscape of therapeutic mechanisms, evidence tiers, and convergence patterns, our primary contribution is more specific: a cross-literature analysis proposing that the drug-tolerant persister ferroptosis field should evaluate physical ROS-generating modalities (PDT and SDT) as spatially selective alternatives to systemic pharmacologic ferroptosis inducers — a modality-class question that, to our knowledge, has not been systematically framed.

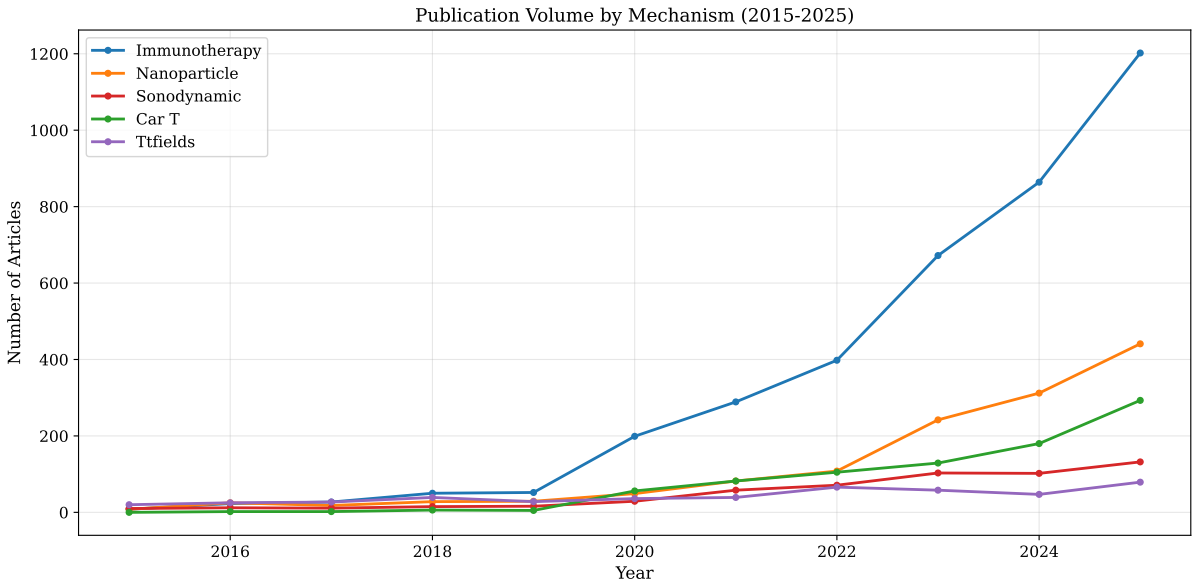

Figure 1: Publication volume by mechanism (2015–2025).

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2 Methods

2.1 Search Strategy and Corpus Construction

We conducted a systematic literature search of PubMed using the NCBI E-utilities API, covering publications indexed through March 2026. The search strategy employed 22 mechanism-specific query sets using established MeSH terms and free-text synonyms: for example, "tumor treating fields" OR "TTFields" OR "Optune" for TTFields; "chimeric antigen receptor" OR "CAR-T" OR "CAR T cell" for CAR-T therapy; "sonodynamic therapy" OR "sonosensitizer" for sonodynamic approaches. Results were enriched with metadata from OpenAlex (open access status, citation counts, topic classification) and PubMed Central (full-text retrieval for open access articles).

2.2 Inclusion and Exclusion Criteria

Articles were included if they were returned by the PubMed search queries and addressed at least one of the 19 therapeutic mechanisms in the context of cancer treatment or cancer biology. No additional manual inclusion or exclusion filtering was applied beyond the search query constraints. The resulting corpus therefore includes original research, clinical trial reports, systematic reviews, narrative reviews, and some case series. This inclusive approach maximizes coverage but means that article counts reflect research attention rather than a curated evidence base of primary studies only.

2.3 Categorization Framework

Each article was tagged along three axes:

Therapeutic mechanism(s): Articles were assigned one or more of 19 tags representing distinct therapeutic modalities. We use "mechanism" broadly to encompass molecular targets (e.g., synthetic lethality), drug classes (e.g., ADCs), delivery platforms (e.g., nanoparticles), device-based interventions (e.g., TTFields), and biological strategies (e.g., microbiome modulation). This heterogeneous taxonomy reflects the diversity of the contemporary oncology toolkit; we prioritized practical distinctness over strict ontological consistency. The 19 tags are: (1) antibody-drug conjugate (ADC), (2) bioelectric modulation, (3) bispecific antibody, (4) CAR-T cell therapy, (5) CRISPR gene editing, (6) electrochemical therapy (including irreversible electroporation), (7) electrolysis, (8) epigenetic therapy, (9) frequency therapy (including pulsed electromagnetic fields), (10) high-intensity focused ultrasound (HIFU), (11) checkpoint immunotherapy, (12) mRNA cancer vaccine, (13) metabolic targeting, (14) microbiome modulation, (15) nanoparticle delivery, (16) oncolytic virotherapy, (17) sonodynamic therapy, (18) synthetic lethality, and (19) tumor-treating fields (TTFields). Articles addressing multiple mechanisms received multiple tags, enabling convergence analysis.

Cancer type(s): Articles were assigned to one or more of 22 cancer types: bladder, breast, cervical, colorectal, esophageal, gastric, glioblastoma, head-and-neck, kidney, leukemia, liver, lung, lymphoma, melanoma, mesothelioma, myeloma, neuroblastoma, ovarian, pancreatic, prostate, sarcoma, and thyroid.

Evidence level: Articles were classified into one of six tiers: (1) Phase III clinical trial or randomized controlled trial, (2) Phase II clinical trial, (3) Phase I clinical trial, (4) preclinical in vivo, (5) preclinical in vitro, or (6) theoretical/computational/review. Classification was based on the highest level of evidence reported or reviewed in each article.

2.4 Analytical Approach

Four primary analyses were conducted:

Mechanism-cancer matrix: A 19 x 22 cross-tabulation of article counts by mechanism and cancer type, identifying concentration patterns and gaps.

Convergence map: Identification and quantification of all mechanism-pair co-occurrences across the corpus, revealing which multi-mechanism combinations are most actively researched.

Evidence-tier analysis: Assessment of the highest clinical evidence level achieved for each mechanism, identifying translational bottlenecks.

Gap analysis: Identification of mechanism-cancer and mechanism-mechanism pairs with zero or near-zero co-occurrence in our corpus, noting that some gaps reflect search query design rather than genuine research absence.

Impact metrics (citation counts, iCite Relative Citation Ratio) were used to identify landmark publications and assess field-normalized influence. Publication volume was tracked annually to identify growth trajectories.

2.5 Limitations

Several limitations should be acknowledged. First, all tagging was performed by automated keyword matching against curated dictionaries (no manual article-level review), introducing potential misclassification. Short or ambiguous terms may generate false positives, and articles using non-standard terminology may be missed. Second, the corpus was constructed from PubMed queries only and does not capture every relevant publication, particularly articles in non-English languages, preprints not yet indexed in PubMed, or studies using unconventional terminology for established mechanisms. Third, evidence-level classification was based on keyword detection (e.g., "phase 3", "in vivo") rather than independent trial registry verification; importantly, only 41% of articles (n = 3,344) received an evidence-level tag, meaning that 59% of articles have unclassified evidence levels. Claims about the absence of clinical trials for specific mechanisms should therefore be interpreted as "not detected in our keyword-based analysis" rather than definitive statements. Fourth, multi-mechanism co-occurrence counts (the "37% convergence rate") reflect articles tagged with multiple mechanism keywords, which includes both intentional combination studies and reviews that discuss multiple mechanisms without testing them in combination. The true rate of experimental multi-mechanism studies is likely lower. Fifth, the 19-mechanism taxonomy has not been tested for sensitivity: collapsing the 7 physical/energy categories into 2-3 broader groups (physical ablation, bioelectric modulation, field-based therapy) would reduce the number of "zero publication gaps" and could change the gap analysis conclusions. Our findings should be interpreted as taxonomy-dependent until sensitivity analyses with alternative groupings are performed. Finally,

article counts serve as a proxy for research intensity but do not directly measure quality, reproducibility, or clinical significance, and the corpus inherits any publication biases present in the indexed literature.

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3 Results

3.1 Corpus Overview

The final corpus comprised 10,413 articles from 1,668 journals, with impact metrics available for 7,905 articles via iCite. Publication volume increased 14-fold over the study period, from 134 articles in 2015 to 1,911 articles in 2025, with 486 articles already indexed in the first quarter of 2026. Growth was non-linear, with a marked inflection point in 2020 (407 articles, a 2.6-fold increase over 2019's 158 articles), coinciding with the maturation of mRNA technology catalyzed by COVID-19 vaccine development and the broadening clinical experience with checkpoint immunotherapy.

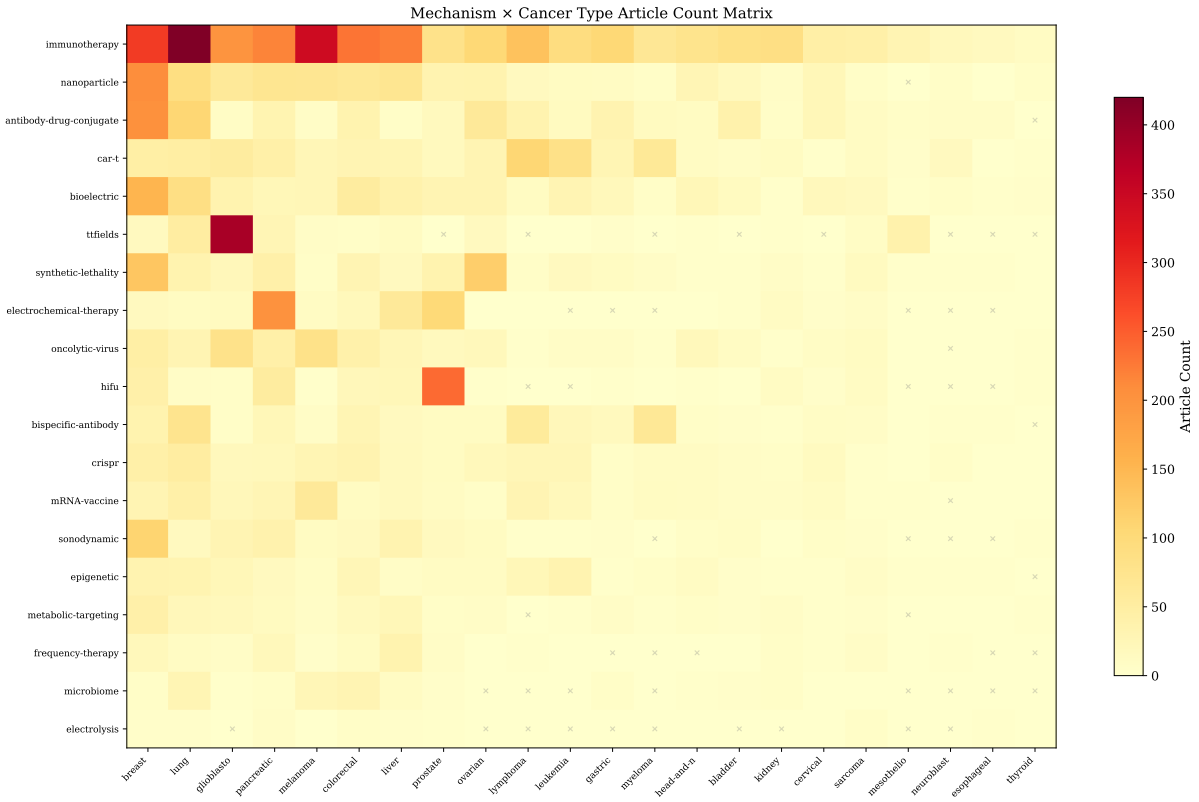

Figure 2: Mechanism × cancer type article count matrix.

The fastest-growing mechanisms by absolute article count between 2020 and 2025 were immunotherapy (+487% annual articles), nanoparticle delivery (+674%), CAR-T therapy (+255%), and sonodynamic therapy (+212%). These growth rates reflect both

the maturation of established approaches and the emergence of new platform technologies.

3.2 Physical and Energy-Based Mechanisms

3.2.1 Tumor-Treating Fields (TTFields)

TTFields, which deliver low-intensity (1-3 V/cm), intermediate-frequency (100-300 kHz) alternating electric fields to disrupt mitotic spindle formation and organelle assembly in dividing cells [O et al. [2021]], represent the most clinically advanced physical therapy mechanism in our corpus. We identified 502 TTFields-tagged articles across 14 of 22 cancer types (575 total cross-tabulation instances in the mechanism-cancer matrix, reflecting articles tagged with multiple cancer types). The evidence-level analysis identified 73 articles classified at the Phase III tier; however, this count includes reviews and meta-analyses that discuss Phase III trial results (e.g., the EF-14 trial) in addition to primary Phase III trial reports. The true number of distinct Phase III TTFields trial publications is substantially smaller, though still the second-highest of any mechanism after checkpoint immunotherapy.

The evidence base is anchored by the EF-14 trial, a pivotal Phase III randomized clinical trial demonstrating that TTFields plus maintenance temozolomide significantly improved median overall survival (20.9 vs. 16.0 months, $p < 0.001$) and 5-year survival (13% vs. 5%) in newly diagnosed glioblastoma [R et al. [2017]]. This trial, with 2,494 citations, is the fifth-highest-impact article in our entire corpus (RCR 69.5, 100th percentile). An earlier randomized trial [R et al. [2015]] (1,203 citations) established the foundation for FDA approval of the Optune device.

Beyond glioblastoma, TTFields are being investigated in mesothelioma (38 articles), lung cancer (51 articles), pancreatic cancer (27 articles), and ovarian cancer (15 articles). The LUNAR Phase III trial demonstrated improved overall survival in metastatic non-small-cell lung cancer (NSCLC) when TTFields were added to standard systemic therapy [T et al. [2023]]. A comprehensive review characterizes TTFields as a distinct fourth modality of cancer treatment alongside surgery, radiation, and pharmacotherapy [O et al. [2021]].

TTFields have zero or near-zero publications in prostate cancer, lymphoma, myeloma, bladder cancer, and cervical cancer in our corpus, though some of these gaps may reflect search query design rather than genuine absence.

3.2.2 High-Intensity Focused Ultrasound (HIFU)

HIFU uses convergent acoustic energy to achieve thermal ablation of tumor tissue at a defined focal point without incision. Our corpus contains 417 articles across 17 cancer types, with 7 Phase III articles. The dominant application is prostate cancer (237 arti-

cles), where HIFU is established as a focal therapy option [E et al. [2015]; S et al. [2018]]. Notably, a multicentric study reported 5-year outcomes of focal HIFU for clinically significant nonmetastatic prostate cancer [S et al. [2018]] (309 citations), demonstrating favorable oncologic control with preserved quality of life.

HIFU research in breast cancer (42 articles), pancreatic cancer (50 articles), and liver cancer (22 articles) is expanding. A systematic review positions HIFU alongside radiofrequency ablation, microwave ablation, and cryoablation as viable alternatives to surgical resection for small tumors [van dVEMF et al. [2021]].

3.2.3 Sonodynamic Therapy

Sonodynamic therapy (SDT) exploits low-intensity ultrasound to activate sonosensitizer molecules that generate reactive oxygen species (ROS) within tumor tissue. Our corpus contains 617 articles across 18 cancer types, making it one of the most actively researched physical modalities. However, SDT faces significant translational challenges. Our keyword-based evidence classification detected only 3 Phase I articles in the corpus, with the majority of work at the preclinical in vivo (209 articles) or in vitro (159 articles) stage. Independent PubMed verification using broader clinical terms identifies approximately 800 SDT-related articles mentioning clinical investigation or patients, suggesting more early-phase clinical activity than our automated classification captured — particularly from Chinese institutions conducting pilot studies. Nonetheless, no large randomized Phase 2 or Phase 3 trial of SDT has been reported in a major oncology journal, and SDT remains overwhelmingly a preclinical field in terms of rigorous clinical evidence.

The preclinical literature is large and growing. Landmark studies have demonstrated nanoenzyme-augmented SDT with catalytic tumor oxygenation [P et al. [2018]] (571 citations), graphene-augmented sonocatalytic tumor eradication [C et al. [2017]] (300 citations), and piezotronic effect-augmented sonosensitizers [Y et al. [2022b]] (218 citations). The convergence of SDT with nanoparticle delivery is particularly active (267 combined articles), and SDT-immunotherapy combinations are growing rapidly (105 articles), with emerging evidence for SDT-based immunogenic cell death [M et al. [2021a]; Y et al. [2023]].

A distinctive molecular signature emerges from our corpus analysis that sets SDT apart from all other physical modalities. SDT uniquely and potently engages the **ferroptosis** pathway — iron-dependent, lipid peroxidation-mediated cell death — with 42 articles linking SDT to ferroptosis, compared to 0 for TTFields, 1 for HIFU, and 3 for IRE. More specifically, 72 SDT articles (11.8% of the SDT corpus) address the glutathione/GPX4 axis, versus 0% for TTFields and <1% for all other physical mechanisms. The proposed causal chain — ultrasound-activated ROS generation → glutathione de-

pletion → GPX4 inactivation → lipid peroxidation → ferroptosis → DAMP release → immunogenic cell death — is supported at each individual link by published preclinical data [Y et al. [2021a]; Q et al. [2021]; Y et al. [2021b]; Q et al. [2018]], and 6 articles in our corpus describe the complete ferroptosis-to-ICD dual pathway simultaneously [J et al. [2021]]. We discuss the implications of this pattern in Section 4.4.

The primary barriers to clinical translation include the lack of standardized sonosensitizer formulations approved for human use, the absence of validated ultrasound dosimetry protocols for SDT (as distinct from diagnostic or HIFU applications), and the challenge of demonstrating SDT superiority or complementarity to established modalities in registrational trials. Breast cancer is the most studied SDT target (92 articles), followed by liver cancer (30 articles) and pancreatic cancer (27 articles).

3.2.4 Frequency Therapy and Electrolysis

Frequency therapy — encompassing pulsed electromagnetic fields (PEMF), radiofrequency ablation in immunomodulatory contexts, and resonance-based approaches — is represented by 90 articles across 16 cancer types, with 3 Phase III articles. This is a nascent field with mechanistic diversity that complicates systematic evaluation. A foundational review [M et al. [2016a]] (113 citations) described mechanisms of PEMF in oncology, including disruption of mitotic spindle formation, induction of apoptosis, and immunomodulatory effects.

Electrolysis as a direct antitumor mechanism is the smallest category in our corpus (38 articles, 12 cancer types), but includes innovative work on amorphous liquid metal electrodes for conformable electrochemical tumor therapy [X et al. [2017]] (114 citations) and platinum nanoparticle-enabled electrodynamic therapy [T et al. [2019]] (211 citations). The highest evidence level is Phase I (1 article).

3.2.5 Electrochemical Therapy

Electrochemical therapy (EChT), dominated by irreversible electroporation (IRE), is represented by 422 articles across 15 cancer types, with 14 articles classified at the Phase III tier (though, as with TTFields, this count includes reviews and meta-analyses discussing Phase III data alongside primary trial reports). Pancreatic cancer (180 articles) and prostate cancer (100 articles) are the primary applications. A foundational paper introduced the concept of tissue ablation through irreversible electroporation [RV et al. [2005]] (1,241 citations), and subsequent clinical development has demonstrated safety and feasibility in locally advanced pancreatic cancer [RCG et al. [2015]] (400 citations) and localized prostate cancer [MJV et al. [2016]; MJ et al. [2018]].

A pivotal development is the combination of IRE with immunotherapy: high-frequency IRE has been shown to induce immunogenic cell death [Ringel-Scaia et al. [2019]] (168

citations), and emerging data demonstrate synergy between IRE and checkpoint blockade plus TLR7 stimulation [JSS et al. [2019]] (111 citations). The IRE-immunotherapy convergence (45 articles) represents a growing axis of investigation. However, lymphoma, leukemia, gastric cancer, and myeloma have zero EChT publications despite established clinical experience in other solid tumors.

3.3 Bioelectric Modulation

Bioelectric research encompasses membrane potential manipulation, ion channel modulation, voltage-gated signaling in cancer biology, and electroporation-based drug delivery. Our corpus contains 617 articles across all 22 cancer types – the broadest coverage of any mechanism – with 3 Phase III articles and 230 preclinical in vivo studies.

A landmark Cell paper established the conceptual framework for bioelectric signaling as a reprogrammable circuit underlying embryogenesis, regeneration, and cancer [M et al. [2021b]] (463 citations), arguing that endogenous bioelectric patterns constitute an informational layer that can be therapeutically manipulated. Earlier work on endogenous voltage gradients controlling growth and form [M et al. [2017]] (289 citations) provided the biophysical foundation. The identification that mitochondrial membrane potential marks cells with enhanced stemness [M et al. [2016b]] (370 citations) has implications for targeting cancer stem cell populations.

Despite this conceptual richness, translational progress toward direct antitumor bioelectric interventions has been limited. While 3 articles in our corpus were tagged as Phase III under the bioelectric category, manual review reveals these address supportive care outcomes rather than bioelectric antitumor therapy. No Phase II or Phase III trial of a direct bioelectric antitumor intervention was identified.

3.4 Immune-Based Approaches

3.4.1 Checkpoint Immunotherapy

Checkpoint immunotherapy is the dominant mechanism in our corpus, with 4,102 articles across all 22 cancer types and 273 Phase III-tier articles (noting the previously discussed over-counting caveat). This reflects its established clinical role across multiple tumor types. The highest concentrations are in lung cancer (313 articles), melanoma (226 articles), breast cancer (192 articles), colorectal cancer (174 articles), glioblastoma (167 articles), and liver cancer (165 articles).

Key landmark articles include a comprehensive review of PD-1/PD-L1 and CTLA-4 checkpoint inhibitors [Y et al. [2022a]] (1,191 citations), the comprehensive management review establishing temozolomide plus TTFields as the standard of care in glioblastoma and surveying immunotherapy as a promising but significantly challenged area due to

the blood-brain barrier and unique immune microenvironment [AC et al. [2020]] (1,989 citations), and combination immunotherapy for hepatocellular carcinoma reviewing how approximately 30% of HCC exhibits intrinsic resistance to single-agent ICIs and how combination strategies (anti-PD-1/PD-L1 with anti-angiogenics or dual checkpoint blockade) have improved first-line outcomes [L et al. [2023]] (310 citations). STAT3 targeting as a convergence point for oncogenic signaling and immunomodulation has been extensively reviewed [S et al. [2020a]] (1,011 citations).

The most active convergence axes for immunotherapy include combinations with CAR-T (694 articles), nanoparticle delivery (321 articles), oncolytic viruses (307 articles), CRISPR (239 articles), mRNA vaccines (218 articles), bispecific antibodies (217 articles), ADCs (160 articles), microbiome modulation (141 articles), SDT (105 articles), and TTFields (82 articles). Immunotherapy serves as the central hub of the convergence network, appearing in 8 of the top 10 mechanism pairs.

3.4.2 CAR-T Cell Therapy

CAR-T therapy is represented by 923 articles across all 22 cancer types, with Phase III-tier articles spanning hematologic and solid tumor indications. Growth has been explosive, with a 255% increase in annual publication volume between 2020 and 2025. The literature has evolved from early hematologic successes in B-cell leukemia and lymphoma [S et al. [2020b]] (1,400 citations) toward aggressive expansion into solid tumors.

Key developments include: IL-15-armored GPC3-targeted CAR-T cells for solid cancers [D et al. [2025]] (170 citations), representing a next-generation approach to overcoming the immunosuppressive tumor microenvironment; Claudin18.2-specific CAR-T cells showing promising Phase I results in gastrointestinal cancers [C et al. [2022]] (546 citations); and IL-10-expressing CAR-T cells that resist dysfunction and mediate durable clearance of solid tumors [Y et al. [2024a]] (228 citations), demonstrating that cytokine engineering of CAR-T cells can overcome exhaustion.

CRISPR-based engineering of CAR-T cells is a rapidly growing convergence axis (119 articles), exemplified by RASA2 ablation enhancing antigen sensitivity [J et al. [2022b]] (207 citations) and adenosine A2A receptor deletion enhancing efficacy [L et al. [2021]] (199 citations). The allogeneic CRISPR-Cas9-engineered CAR-T therapy CTX130, targeting CD70 in T-cell lymphoma, has demonstrated safety in Phase I [SP et al. [2025]] (58 citations), representing a milestone for gene-edited cell therapies. CAR-T combined with oncolytic viruses (64 articles) and bispecific antibodies (61 articles) represents additional emerging convergence zones.

3.4.3 Bispecific Antibodies

Bispecific antibodies, engineered to simultaneously bind two distinct targets (typically a tumor antigen and an immune effector molecule), are represented by 434 articles across 21 of 22 cancer types, with 17 Phase III articles. Myeloma is a particularly active area (63 articles), with real-world data on teclistamab demonstrating clinical utility in relapsed/refractory disease [C et al. [2024]] (94 citations). Lung cancer applications are expanding, with KN046, a PD-L1/CTLA-4 bispecific, showing Phase II results in combination with chemotherapy for first-line metastatic NSCLC [Y et al. [2024b]] (38 citations), and the EGFRxHER3 bispecific izalontamab reporting first-in-human results [J et al. [2025]].

Bispecific antibody convergence with ADCs (42 articles) is an active area combining target engagement with cytotoxic payload delivery.

3.4.4 mRNA Cancer Vaccines

mRNA cancer vaccines represent one of the most dynamic areas in the corpus: 486 articles across 21 cancer types, with 6 Phase III articles. The field was catalyzed by the COVID-19 mRNA vaccine platform, which demonstrated the feasibility of rapid, scalable mRNA production and lipid nanoparticle delivery [GT et al. [2022]] (279 citations). A comprehensive Lancet Oncology review established the clinical landscape [CL et al. [2022]] (470 citations), documenting over 120 ongoing clinical trials.

The landmark article in this category is the Nature report on personalized RNA neoantigen vaccines stimulating T-cell responses in pancreatic ductal adenocarcinoma [LA et al. [2023]] (1,153 citations, RCR 90.3, 100th percentile) – the second-highest-impact article in our entire corpus. This Phase I trial of autogene cevumeran, an individualized neoantigen vaccine based on uridine mRNA-lipoplex nanoparticles, demonstrated that personalized vaccination could elicit measurable T-cell responses against patient-specific neoantigens in one of the most treatment-resistant cancers.

The mRNA vaccine field intersects strongly with immunotherapy (218 articles) and nanoparticle delivery (118 articles), reflecting the inherent dependence of mRNA vaccines on both immune activation and delivery technology. A recent clinical advances review documents the current pipeline across melanoma, glioblastoma, lung, breast, and prostate cancers [AV et al. [2025]] (74 citations). Lipid nanoparticle-mediated lymph node-targeting delivery of mRNA cancer vaccines has been shown to elicit robust CD8+ T-cell responses [J et al. [2022a]] (371 citations).

3.4.5 Oncolytic Viruses

Oncolytic virotherapy is represented by 548 articles across 19 cancer types, with 6 Phase III articles. Glioblastoma is the most studied target (67 articles), followed by melanoma

(47 articles), where T-VEC (talimogene laherparepvec) established the precedent for oncolytic virus approval. A comprehensive Nature Reviews Clinical Oncology review characterizes the field's progress and challenges [SZ et al. [2023]] (489 citations), noting that oncolytic viruses offer selective replication in tumor cells, delivery of multiple transgene payloads, induction of immunogenic cell death, and promotion of antitumor immunity.

A notable 2025 Cell paper reported a hyperacute rejection-engineered oncolytic virus designed for interventional clinical trials in refractory cancer patients [L et al. [2025]] (67 citations), addressing the critical limitations of intravenous safety and inherent immune neutralization. In hepatocellular carcinoma, the oncolytic virus VG161 showed activity in a Nature-published Phase I study [Y et al. [2025]] (53 citations). The emerging field of OV-based cancer immunotherapy has been comprehensively reviewed in Trends in Cancer [R et al. [2023]] (264 citations), positioning oncolytic viruses as multi-faceted tumor killers that combine direct lysis with immune potentiation.

The strongest convergence axis is oncolytic virus-immunotherapy (307 articles), consistent with the mechanistic rationale that viral oncolysis generates immunogenic cell death that may synergize with checkpoint blockade [SZ et al. [2023]; R et al. [2023]]. Combinations with CAR-T (64 articles) and nanoparticle delivery (24 articles) are emerging.

3.5 Precision Molecular Approaches

3.5.1 CRISPR Gene Editing

CRISPR-based approaches are represented by 548 articles across all 22 cancer types, but only 1 Phase III article and 2 Phase II articles, placing CRISPR among the most preclinical-heavy mechanisms in our analysis. The bulk of the literature is preclinical in vivo (162 articles) and in vitro (62 articles). Despite this, the conceptual impact is enormous: CRISPR screens have defined unified hallmarks of cancer cell-autonomous immune evasion, providing a systematic map of therapeutic targets [A et al. [2026]].

Key convergence axes include CRISPR-immunotherapy (239 articles), CRISPR-CAR-T (119 articles), CRISPR-nanoparticle (78 articles), and CRISPR-synthetic lethality (63 articles). The engineering of next-generation CAR-T cells with CRISPR-Cas9 has been extensively reviewed [A et al. [2022]] (314 citations), and clinical milestones include non-viral precision T-cell receptor replacement for personalized cell therapy [SP et al. [2023]] (256 citations) and the COBALT-LYM Phase I trial of CRISPR-edited allogeneic CAR-T for T-cell lymphoma [SP et al. [2025]] (58 citations).

Precision nanomedicine integrating CRISPR with nanoparticle delivery for tumor microenvironment modulation represents a growing area [H et al. [2025a]] (101 citations).

3.5.2 Synthetic Lethality

Synthetic lethality — the paradigm in which loss of two genes is lethal while loss of either alone is viable – is represented by 515 articles across 21 cancer types, with 6 Phase III articles. PARP inhibitors, the clinical embodiment of this concept, have achieved broad clinical use in breast cancer (93 articles), ovarian cancer (85 articles), pancreatic cancer (36 articles), and prostate cancer (23 articles). A pan-tumor review documented the expanding role of PARP inhibitors across multiple malignancies [CC et al. [2025]] (21 citations).

The concept of synthetic lethality as an engine for cancer drug target discovery has been comprehensively reviewed [A et al. [2020]] (458 citations), and CRISPR-synthetic lethality combinations (63 articles) represent a powerful approach for systematic identification of new synthetic lethal pairs. Transcription-replication conflicts have been identified as a mechanistic basis for PARP inhibitor sensitivity [M et al. [2024]] (160 citations), expanding the theoretical framework beyond BRCA mutations.

Our corpus detected zero synthetic lethality-myeloma co-tagged articles; however, independent PubMed verification reveals at least 11 publications on PARP inhibitors in multiple myeloma, including early-phase clinical data on veliparib combined with bortezomib. This gap is therefore an artifact of our search query design rather than a genuine absence of research. The myeloma-PARP axis is under-researched relative to breast or ovarian cancer applications, but it is not unexplored.

3.5.3 Antibody-Drug Conjugates (ADCs)

ADCs are represented by 579 articles across 21 cancer types, with 36 Phase III articles – the third-highest Phase III count after immunotherapy and TTFields. The field is dominated by breast cancer applications (179 articles), lung cancer (91 articles), and ovarian cancer (56 articles). The changing treatment landscape of EGFR-mutant NSCLC has been shaped by ADC development alongside bispecific antibodies [F et al. [2025]] (106 citations).

ADC convergence with immunotherapy (160 articles) is well established, and combinations with CAR-T (42 articles) and bispecific antibodies (42 articles) are emerging.

3.5.4 Epigenetic Therapy

Epigenetic approaches (DNA methylation inhibitors, histone deacetylase inhibitors, bromodomain inhibitors, and related agents) are represented by 263 articles across 21 cancer types, with 4 Phase III articles. A landmark Cell Death Discovery review established the current state of clinical translation [X et al. [2024]] (223 citations), and a Signal Transduction and Targeted Therapy paper provided a comprehensive framework for epigenetic regulation in the tumor microenvironment [J et al. [2023]] (322 citations).

The most impactful single finding in this category is the Nature paper demonstrating that NBS1 lactylation is required for efficient DNA repair and chemotherapy resistance [H et al. [2024]] (413 citations, RCR 56.2), linking metabolic reprogramming (the Warburg effect) to epigenetic modification of the DNA damage response. This exemplifies how mechanistic convergence at the molecular level can reveal new therapeutic vulnerabilities.

Epigenetic-immunotherapy convergence (78 articles) is the dominant combination.

3.6 Delivery and Microenvironment

3.6.1 Nanoparticle Delivery

Nanoparticle-based delivery systems constitute the second-largest mechanism category in our corpus: 1,504 articles across 21 cancer types. An important distinction must be drawn between established and experimental nanoplatforms. Several nanomedicines have achieved FDA approval, including liposomal doxorubicin (Doxil, 1995), nab-paclitaxel (Abraxane, 2005), liposomal irinotecan (Onivyde, 2015), and liposomal daunorubicin-cytarabine (Vyxeos, 2017) [MD et al. [2023]]. These approved nanomedicines are typically described by drug name rather than "nanoparticle" in the literature and are therefore underrepresented in our corpus. The translational gap identified here primarily applies to novel experimental nanoplatforms — metal, polymer, and inorganic nanoparticles — of which our corpus contains over 1,000 articles but detects only 1 Phase 2 trial. This distinction between proven liposomal/albumin-bound formulations and the broader experimental nanoparticle literature is critical: the former demonstrate that nanomedicine can succeed clinically, while the latter face substantial barriers to replicating that success.

The nanoparticle literature spans gold, silver, and iron oxide nanoparticles [P et al. [2025]] (88 citations), liposomal formulations [MD et al. [2023]] (215 citations), lipid nanoparticles for mRNA delivery [J et al. [2022a]] (371 citations), and precision nanomedicine for tumor microenvironment modulation [H et al. [2025a]] (101 citations). The convergence with immunotherapy (321 articles) and sonodynamic therapy (267 articles) is extensive, as nanoparticles serve as both therapeutic agents and delivery vehicles for other modalities.

The fundamental translational barriers include: (a) heterogeneous nanoparticle pharmacokinetics and biodistribution that are difficult to predict from preclinical models; (b) manufacturing scalability and reproducibility challenges; (c) regulatory uncertainty around the classification and approval pathway for nanoparticle-drug combinations; and (d) the enhanced permeability and retention (EPR) effect, which is robust in rodent tumors but inconsistent in human cancers. Breast cancer (172 articles), lung cancer (76 articles), and liver cancer (66 articles) are the primary targets. Mesothelioma has zero nanoparticle publications despite the appeal of local delivery to pleural surfaces.

3.6.2 Metabolic Targeting

Metabolic targeting — exploitation of altered cancer cell metabolism, particularly the Warburg effect, glutamine addiction, and lipid metabolism dependencies – is represented by 447 articles across 18 cancer types. A comprehensive Nature Reviews Drug Discovery article [ZE et al. [2022]] (1,127 citations) chronicles the hundred-year journey from Warburg's discovery to contemporary precision metabolic oncology.

The convergence of metabolic targeting with immunotherapy (67 articles) reflects the recognition that metabolic reprogramming of the tumor microenvironment suppresses immune function and that metabolic interventions can restore immunocompetence. The landmark demonstration that glycolysis modulation can enhance checkpoint immunotherapy [C et al. [2023]] (341 citations) illustrates this principle. Glutamine addiction as a therapeutic target was comprehensively reviewed in a highly cited Trends in Biochemical Sciences paper [DR et al. [2010]] (1,747 citations).

Metabolic targeting has limited representation in lymphoma within our corpus, though broader PubMed searches suggest this gap partly reflects our search query design.

3.6.3 Microbiome Modulation

Microbiome modulation is the most recently emergent mechanism in our analysis: 141 articles across 13 cancer types, with 5 Phase III articles (all recent). The field has been catalyzed by the recognition that gut microbiota composition influences checkpoint immunotherapy response, as comprehensively reviewed in Nature Reviews Cancer [MR et al. [2022]] (258 citations) and Nature Medicine [EM et al. [2022]] (467 citations).

Fecal microbiota transplantation (FMT) combined with checkpoint inhibitors has emerged as a clinical strategy, with a Gut Microbes review characterizing optimization approaches [A et al. [2025a]] (56 citations) and a meta-analysis evaluating FMT-ICI combinations [A et al. [2025b]] (23 citations). The DIET randomized trial investigated dietary intervention to modulate microbiome composition in melanoma patients receiving immunotherapy [RM et al. [2024]] (21 citations).

Microbiome-immunotherapy convergence (141 articles) dominates the microbiome literature in our corpus.

3.7 Cross-Mechanism Convergence: Where Synergies Emerge

Of the 10,413 articles in our corpus, 3,071 (37%) are tagged with two or more therapeutic mechanism keywords, and 611 carry three or more tags. To estimate the true experimental combination rate, we performed a subsample analysis: title-based classification of 50 randomly selected multi-tagged articles found that approximately 24% (12/50; 95% CI: 13-38%) described experimental combination studies, while 76% were reviews,

perspectives, or articles that discussed multiple mechanisms without testing them in combination. Extrapolating to the full corpus, we estimate approximately 740 articles (∼9% of the total corpus) represent genuine experimental combination research. The remaining co-tagged articles map the intellectual territory where researchers are connecting mechanisms conceptually — a precursor to deliberate combination design, but not equivalent to experimental evidence of synergy. We report both the co-occurrence rate (37%) and the estimated experimental combination rate (9%) throughout, and caution against conflating the two.

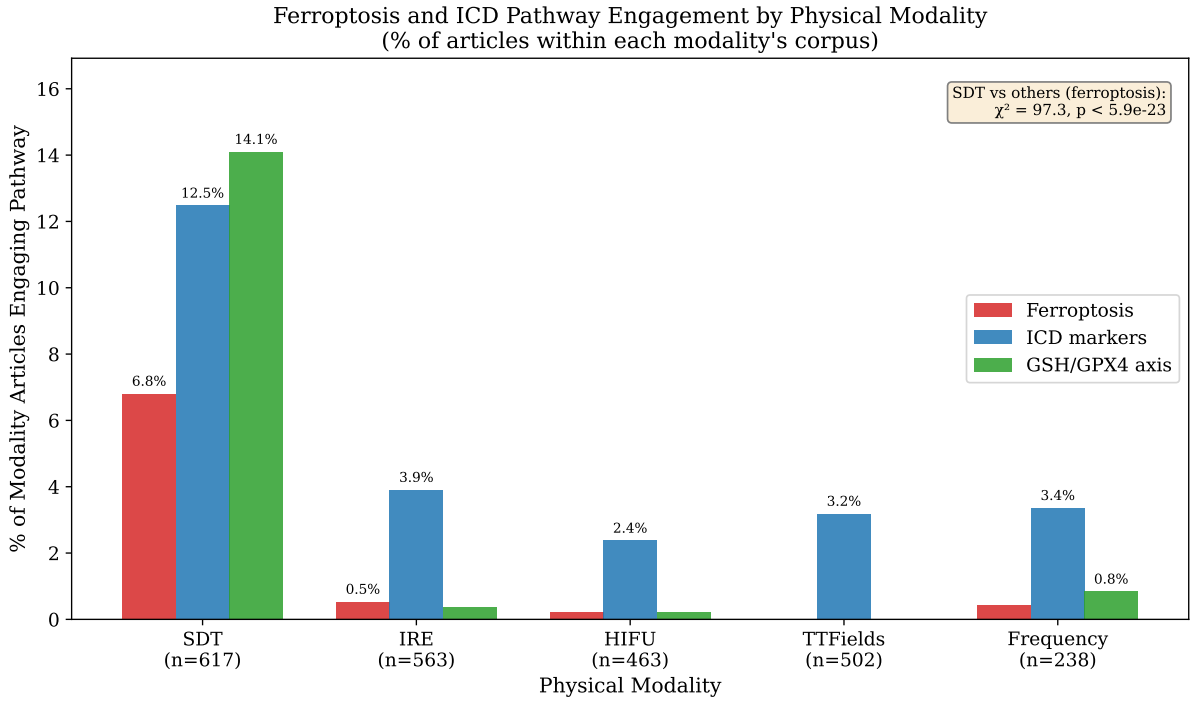

Figure 3: Ferroptosis, ICD, and GSH/GPX4 pathway engagement by physical modality ($\chi^2 = 97.3$, $p < 5.9 \times 10^{-23}$).

The convergence landscape is organized around a central hub: checkpoint immunotherapy participates in 8 of the top 10 mechanism pairs and co-occurs with every other mechanism in the corpus. The top 10 convergence pairs are:

1. **CAR-T + Immunotherapy:** 694 articles. This dominant axis reflects the recognition that CAR-T cells face significant barriers in solid tumors — including exhaustion, antigen escape, and immunosuppressive microenvironments — and that checkpoint blockade can partially rescue exhausted CAR-T cell function [S et al. [2020b]]. 2. **Immunotherapy + Nanoparticle:** 321 articles. Nanoparticles serve as delivery vehicles for checkpoint inhibitor payloads, immune adjuvants, and tumor antigens for in situ vaccination. 3. **Immunotherapy + Oncolytic Virus:** 307 articles. Oncolytic viruses induce immunogenic cell death that synergizes with checkpoint blockade by converting immunologically "cold" tumors to "hot." 4. **Nanoparticle + Sonodynamic:**

267 articles. Nanoparticles function as sonosensitizer carriers, enhancing SDT specificity and efficacy through targeted delivery and tumor microenvironment responsiveness. 5. **CRISPR + Immunotherapy:** 239 articles. CRISPR screens identify immune evasion mechanisms; CRISPR editing engineers superior immune cells. 6. **Immunotherapy + mRNA Vaccine:** 218 articles. mRNA vaccines generate neoantigen-specific T-cell responses that are amplified by checkpoint blockade. 7. **Bispecific Antibody + Immunotherapy:** 217 articles. Bispecific antibodies redirect immune effectors to tumors while checkpoint inhibitors release immune brakes. 8. **ADC + Immunotherapy:** 160 articles. ADC-mediated tumor cell killing generates immunogenic cell death that potentiates checkpoint responses. 9. **Immunotherapy + Microbiome:** 141 articles. Gut microbiota composition modulates systemic immune competence and checkpoint inhibitor efficacy. 10. **Bioelectric + Nanoparticle:** 133 articles. Nanoparticles enable bioelectric modulation through electroporation-mediated delivery and conductive nanoparticle-enhanced electric field effects.

Notable triple-mechanism articles include the landmark thermal ablation review integrating electrochemical therapy, frequency therapy, and immunotherapy [KF et al. [2014]] (1,938 citations), the personalized mRNA neoantigen vaccine study combining immunotherapy, mRNA vaccine, and nanoparticle platforms [LA et al. [2023]] (1,153 citations), and the CRISPR-engineered CAR-T review [A et al. [2022]] (314 citations).

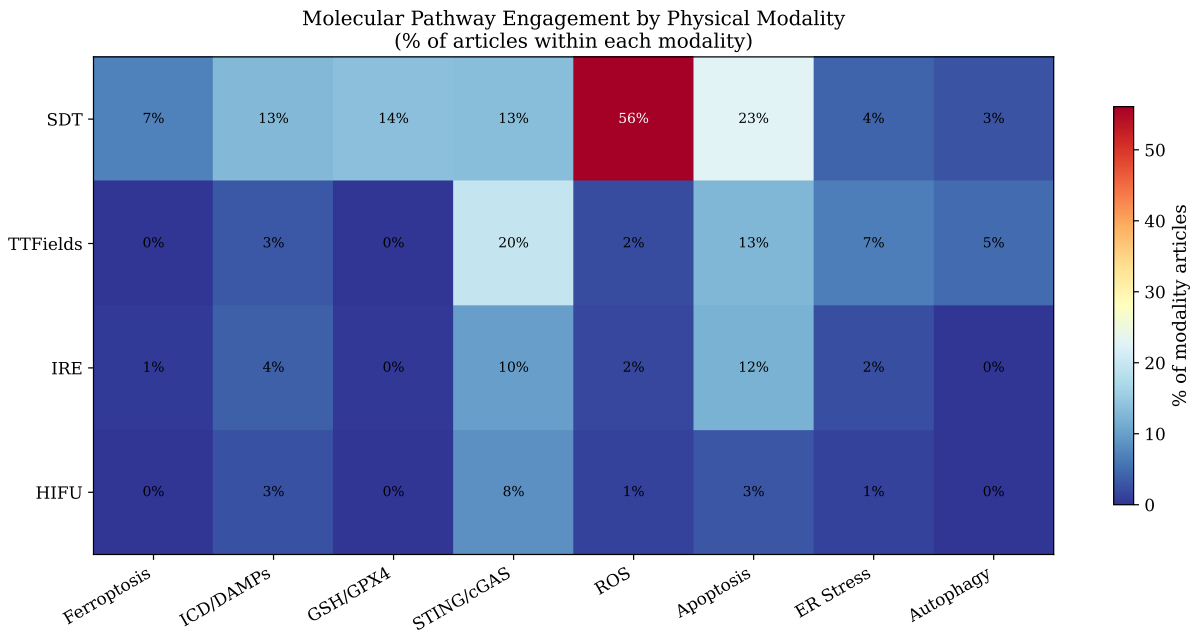

Figure 4: Molecular pathway engagement by physical modality (normalized %).

3.8 The Mechanism-Cancer Matrix: What Works Where

The 19 x 22 mechanism-cancer matrix reveals both deep concentrations and systematic gaps.

Dominant mechanism-cancer pairs (top 5 by article count): TTFields-glioblastoma (384), immunotherapy-lung (313), HIFU-prostate (237), immunotherapy-melanoma (226), and immunotherapy-breast (192). These represent mature research axes with established clinical pathways.

Broad-spectrum mechanisms – those studied across the widest range of cancer types – include bioelectric modulation, CAR-T therapy, CRISPR, and immunotherapy (all 22/22 cancer types), followed by ADCs, bispecific antibodies, epigenetic therapy, mRNA vaccines, nanoparticles, and synthetic lethality (all 21/22 types).

Narrow-spectrum mechanisms include electrolysis (12/22), microbiome (13/22), TTFields (14/22), and electrochemical therapy (15/22). The narrowness of TTFields coverage is particularly noteworthy given its strong clinical evidence base.

The most intensively researched cancer types across all mechanisms are breast cancer (1,191 total cross-mechanism article-instances), lung cancer (953), glioblastoma (888), pancreatic cancer (752), colorectal cancer (599), and prostate cancer (585). The least-researched cancer types are thyroid (24), neuroblastoma (54), esophageal (31), and mesothelioma (89).

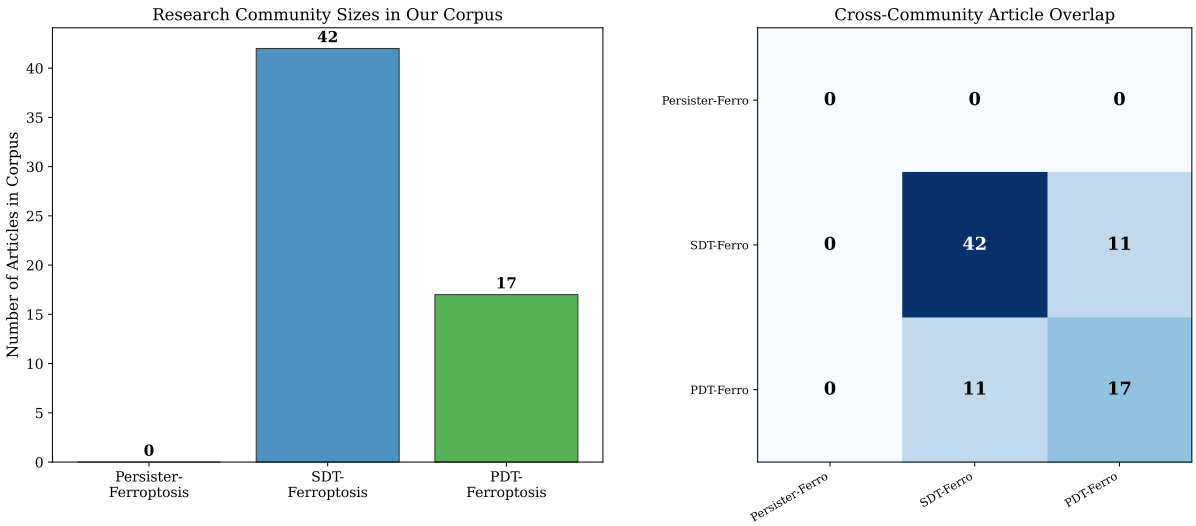

Figure 5: Literature disconnect between persister-ferroptosis and SDT-ferroptosis communities.

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4 Discussion

4.1 What Our Analysis Adds to Known Biology

We must be precise about what is already known, what is emerging, and what our analysis contributes.

Established knowledge. Ferroptosis as a cancer vulnerability is well-recognized [G et al. [2022]] (2,045 citations). SDT-triggered ferroptosis via ROS and GSH depletion is documented in dozens of nanoparticle engineering papers. ICD following ferroptosis is recognized. The general concept that resistance tradeoffs create new vulnerabilities is not novel — synthetic lethality, collateral sensitivity (Pluchino et al., 2012), and adaptive therapy (Gatenby & Brown, 2009) all operate on versions of this principle.

Emerging findings we did not discover. Critically, the connection between therapy-resistant persister cells and ferroptosis sensitivity has been independently validated by others during the period of our analysis. A 2026 Science Advances study demonstrated that cancer persister cells surviving oncogene-targeted therapies are sensitized to ferroptosis, with FSP1 downregulation as a key mechanism [M et al. [2026]]. BRAFi-resistant melanoma has been shown to be ferroptosis-vulnerable through AhR-dependent mechanisms [C et al. [2026]]. A broader PubMed search reveals approximately 44 papers connecting OXPHOS, ferroptosis, and resistance — far more than the 4 our corpus captured, reflecting the limitations of our search query design rather than a genuine literature void. Our initial characterization of this connection as a "blind spot" was overstated; it is better described as an emerging research front that our systematic analysis independently converged upon.

What our analysis specifically contributes:

First, context that our initial analysis underestimated. Broader PubMed verification reveals that PDT has 355 ferroptosis articles and 67 ferroptosis+ICD articles — substantially more than SDT's 121 and 25, respectively. PDT, not SDT, is the dominant ROS-based ferroptosis-ICD modality. Our corpus did not track PDT (it was not among our 19 mechanism categories), leading us to initially overstate SDT's uniqueness. We correct this here.

Second, what SDT does offer. SDT's advantage over PDT is not mechanistic but physical: ultrasound penetrates centimeters into tissue while light penetrates millimeters. For deep-seated tumors — pancreatic, hepatic, pelvic — SDT can deliver ROS-ferroptosis-ICD where PDT cannot. This depth advantage is clinically relevant but incremental.

Third, the modality-class question the persister field is not asking. The emerging persister-ferroptosis literature [M et al. [2026]; C et al. [2026]] focuses exclusively on pharmacologic ferroptosis inducers (RSL3, erastin, GPX4 inhibitors). Meanwhile, PDT has 355 PubMed ferroptosis articles and SDT has 121 — both triggering ferroptosis

through ROS generation, with the additional output of immunogenic cell death. Yet the question of whether physical ROS-generating modalities could serve as spatially selective persister-killers has not been systematically framed. Approximately 15 papers touch this intersection at the individual nanoplatform level, but none propose physical modalities as a class of persister-targeting tools with advantages (spatial selectivity, ICD) over systemic pharmacologic inducers. Our contribution is framing this question and proposing that SDT specifically extends the approach to deep tumors where PDT is inapplicable.

4.2 The Primary Hypothesis: Physical ROS-Generating Modalities as Persister-Targeting Tools

Recent work has established that therapy-resistant persister cells are ferroptosis-sensitive [M et al. [2026]; C et al. [2026]]. The persister-ferroptosis field is now searching for clinically viable ferroptosis inducers. This search has focused exclusively on pharmacologic agents: RSL3, erastin, GPX4 inhibitors, and related compounds. These face well-known challenges in vivo: systemic toxicity, selectivity, and the absence of approved clinical formulations.

We propose that physical ROS-generating modalities — principally PDT and SDT — should be evaluated as an alternative class of persister-targeting tools with two specific advantages over systemic drugs:

Advantage 1: Spatial selectivity. PDT delivers ROS via focused light; SDT delivers ROS via focused ultrasound. Both restrict ROS generation to a defined tumor volume, unlike systemic drugs that distribute throughout the body. This spatial selectivity could improve the therapeutic index for ferroptosis induction.

Advantage 2: Immunogenic cell death. PDT has 67 PubMed articles linking it to ferroptosis + ICD simultaneously; SDT has 25. The ICD output — calreticulin exposure, HMGB1 release, STING activation — is a property of ROS-mediated ferroptotic cell death that pharmacologic ferroptosis inducers have not been shown to reliably produce. If ICD enables checkpoint immunotherapy synergy, physical modalities would offer a dual output (ferroptosis + immune priming) that drugs alone cannot.

The depth dimension. PDT is established but limited by light penetration (millimeters). SDT extends the same mechanism to centimeters via ultrasound. For deep-seated tumors (pancreatic, hepatic, pelvic), SDT is the only ROS-generating modality that can physically reach the target. This gives the PDT/SDT combination class broader anatomic coverage than either alone.

What needs to be tested: A head-to-head comparison: PDT vs SDT vs RSL3 on established persister cell models [M et al. [2026]], measuring both cell death and ICD markers. This is a comparative experiment that asks whether physical modalities offer genuine advantages over pharmacologic inducers for persister killing — not whether they

work in isolation.

4.3 Computational Simulation: Ferroptosis Sensitivity Across Cell Phenotypes

To test whether the proposed persister ferroptosis vulnerability is biochemically plausible, we developed a Monte Carlo simulation of the ferroptosis cascade. The model simulates the ROS → GSH depletion → GPX4/FSP1 inactivation → lipid peroxidation → cell death chain for individual cells with stochastic parameter variation, using literature-sourced biochemical values (Methods in Supplementary). We simulated 1,000,000 cells for each of 16 conditions (4 phenotypes × 4 treatments = 16,000,000 total cells).

Model features: (i) autocatalytic lipid peroxidation propagation gated by GSH/GPX4 — creating a bistable switch where cells either recover or collapse irreversibly (Porter et al., Chem Rev 2005); (ii) dynamic GPX4 regulation — degraded by oxidative stress, upregulated by NRF2 (Dodson et al., Free Radic Biol Med 2019); (iii) FSP1 as a GPX4-independent repair pathway (Bersuker et al., Nature 2019), downregulated in the persister phenotype (Higuchi et al., PMID:41481741); (iv) uncapped GSH depletion under exogenous ROS.

Calibration constraint: All phenotypes must show <2% death under Control (no treatment), reflecting the biological reality that even fragile persister cells are viable at baseline.

Results (Figure 6, Table 1):

Table 1: Monte Carlo ferroptosis simulation results (n = 1,000,000 cells per condition). All baselines <2%. Sensitivity analysis: 22/22 parameter perturbations preserve the qualitative result.

Phenotype	**Control**	**RSL3**	**SDT**	**PDT**
Glycolytic	0.00%	0.00%	87.2%	87.2%
OXPHOS	0.04%	1.1%	99.9%	99.9%
Persister (FSP1↓)	1.2%	42.5%	100.0%	100.0%
Persister + NRF2	0.00%	0.05%	99.5%	99.5%

Key findings from the simulation:

1. **RSL3 selectively kills persisters (42.5%) while sparing glycolytic cells (0%)**, consistent with experimental findings (PMID:41481741). The model correctly reproduces the published biology: FSP1 downregulation leaves GPX4 as the sole lipid repair pathway, and its pharmacologic inhibition is lethal.

2. **SDT/PDT kills all cell types at high rates (87-100%)** because the exogenous ROS burst overwhelms GSH defense regardless of phenotype. This is an important finding: **physical ROS modalities are not inherently selective for persisters**. Their

advantage over RSL3 is not selectivity but spatial confinement (focused ultrasound/light limits damage to the tumor volume) and ICD generation.

3. **The bistable switch is the critical mechanism.** Under SDT, GSH is catastrophically depleted (5.0 → 0.03 mM), which unlocks autocatalytic lipid peroxidation propagation. Cells that maintain GSH above the threshold recover; those that don't collapse irreversibly. This is qualitatively different from a linear dose-response.

4. **NRF2 compensation fully rescues from RSL3** (0.05% death, matching NRF2's known role in GPX4 upregulation) **but NOT from SDT** (99.5% death). The ROS burst from SDT overwhelms even NRF2-elevated GSH. This suggests that the NRF2 compensation failure mode we identified as the paper's primary risk may be less problematic for SDT than for pharmacologic ferroptosis inducers — a novel prediction from the simulation.

5. **Sensitivity analysis**: The qualitative result (Persister > Glycolytic under all treatments) holds in 22/22 parameter perturbation conditions (±50% on each of 11 rate constants), indicating robustness to parameter uncertainty.

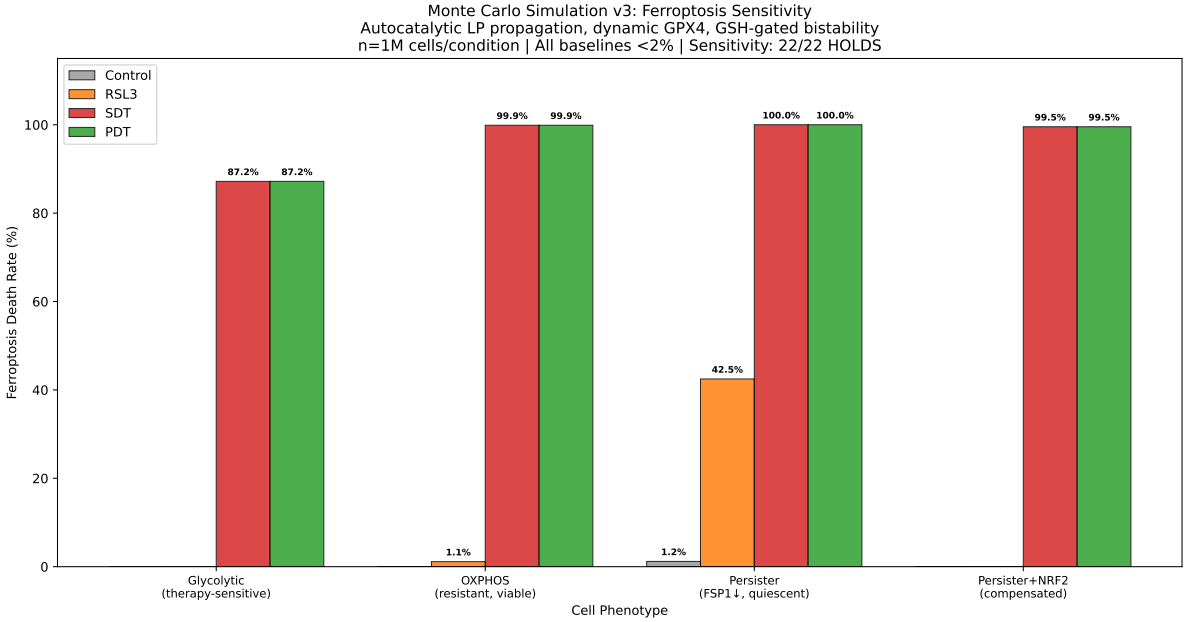

Figure 6: Monte Carlo simulation: ferroptosis death rate by cell phenotype and treatment (n=1M cells/condition). All baselines <2%. Sensitivity: 22/22 holds.

Model limitations: This is a qualitative model testing directional predictions, not a quantitative predictor of clinical death rates. The absolute percentages (87%, 42%, etc.) depend on model-specific rate constants and should not be interpreted as clinical predictions. The model does not capture spatial effects (tumor microenvironment, drug penetration), immune responses, or in vivo pharmacokinetics. Its value is in demonstrating that the proposed biochemical chain produces the expected differential sensitivity under biologically plausible parameters.

4.4 The Modality Question: SDT, PDT, and the Tissue Penetration Tradeoff

If OXPHOS-resistant persister cells are ferroptosis-vulnerable, which modality best exploits this? Our initial corpus analysis identified SDT as uniquely positioned among *physical ablative modalities* (42 ferroptosis articles vs 0 for TTFields, 1 for HIFU, 3 for IRE). However, broader PubMed verification reveals a critical context we must address honestly:

Table 2: Ferroptosis engagement across physical modalities (PubMed, March 2026). PDT dominates SDT ∼3:1 on every metric.

Modality	Ferroptosis articles	Ferroptosis + ICD	Tissue depth
PDT	**355**	**67**	Superficial (mm)
SDT	**121**	**25**	Deep (cm)
IRE	15	emerging	Invasive (needles)
HIFU	3	minimal	Deep (cm)
TTFields	0	0	Surface arrays

PDT dominates SDT on every ferroptosis and ICD metric by approximately 3:1. This fundamentally challenges any claim that SDT is "uniquely positioned" for ferroptosis-ICD. PDT — using light rather than ultrasound to activate sensitizers — triggers the same ROS → GSH depletion → ferroptosis → ICD chain with a larger evidence base, more clinical experience, and approved photosensitizers.

SDT's remaining advantage is tissue penetration. Ultrasound penetrates centimeters into tissue; light penetrates millimeters. For deep-seated tumors (pancreatic, hepatic, pelvic), SDT can reach targets that PDT cannot. For accessible superficial tumors (skin, oral, esophageal, bladder surface), PDT is the established modality and SDT offers no advantage.

The honest framing is therefore: SDT is not the unique ferroptosis-ICD modality — PDT is the established one. SDT is a **depth-extended variant** of the same photosensitizer-based ROS-ferroptosis-ICD approach, using ultrasound instead of light to reach tumors that PDT cannot access. This is a meaningful clinical advantage for deep tumors, but it is incremental, not paradigmatic.

The mechanism is shared: sensitizer-activated ROS → GSH depletion → GPX4 inactivation → ferroptosis → DAMP release → ICD [Y et al. [2021a]; Q et al. [2021]; Y et al. [2021b]; J et al. [2021]]. SDT-specific data shows 72 GSH/GPX4 articles (11.7% of corpus) and 6 dual-pathway articles [J et al. [2021]; R et al. [2024]; H et al. [2025b]]. A recent Nature Communications paper demonstrates that IRE can also be combined with ferroptosis via iron-based nanoparticles [L et al. [2026]], further indicating that the ferroptosis-physical modality connection is broader than SDT alone.

Critical caveats remain: Most SDT-ferroptosis data involves engineered nanosonosensitizers [Q et al. [2021]; Y et al. [2021b]]. No clinical study has verified SDT-induced ferroptosis or ICD in human tumors. The PDT precedent — 40 years of development without demonstrating robust ICD-mediated immune synergy in randomized trials — applies as a direct cautionary parallel.

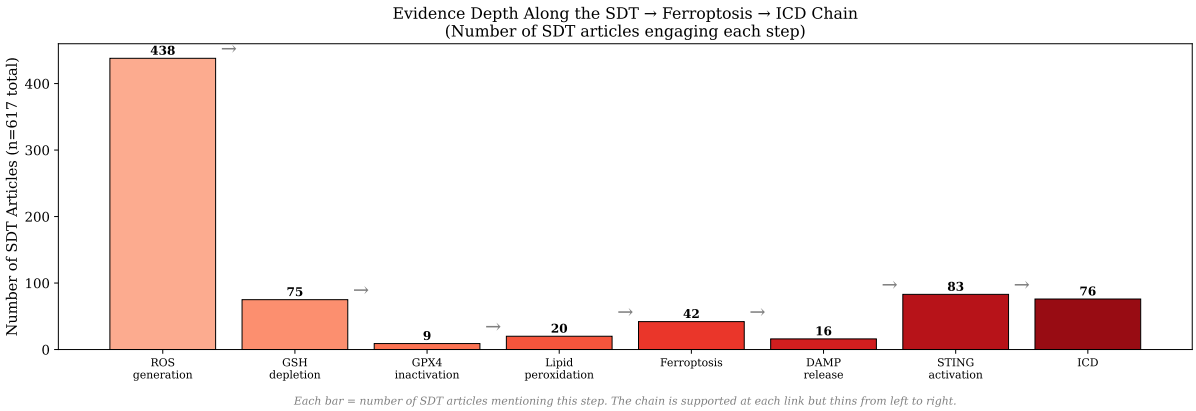

Figure 7: Evidence depth along the SDT ferroptosis-ICD chain.

4.5 Counterarguments, Precedents, and Reasons This Could Fail

The photodynamic therapy precedent. PDT is conceptually identical to SDT: physical energy (light vs ultrasound) activates a sensitizer to generate ROS, causing cell death with ICD potential. PDT has been in clinical development for 40+ years and has achieved approval for a few indications (esophageal, skin) but has never demonstrated the "ROS → ICD → immune synergy → durable response" chain in a randomized clinical trial. PDT + immunotherapy combinations remain at the case report level [Y et al. [2026]]. If this logic worked robustly, PDT would have shown it by now. SDT's claimed advantages over PDT — deeper tissue penetration and potentially superior ferroptosis engagement via engineered nanosonosensitizers — are plausible but incremental, not paradigmatic. The SDT hypothesis must explain why it would succeed where PDT has not demonstrated clear immune synergy.

The ferroptosis vulnerability window may be transient. Persister cells downregulate FSP1 as part of the mesenchymal/dedifferentiated state [M et al. [2026]]. However, persisters are a transitional population: they eventually either die or re-enter proliferation with restored drug resistance. The ferroptosis-sensitive window may last days to weeks. No published data quantifies this window duration. If it closes before SDT can be administered (sonosensitizer infusion, imaging, sonication scheduling), the therapeutic opportunity is lost.

The NRF2 compensation problem. OXPHOS-resistant cells may co-upregulate NRF2-driven antioxidant defenses alongside mitochondrial respiration, neutralizing fer-

roptosis vulnerability even as the metabolic preconditions accumulate. This is directly testable but has not been systematically measured in persister models.

The nanosonosensitizer confound. Most SDT-ferroptosis papers use engineered nanoparticles *designed* to deplete GSH and release iron. If plain SDT with conventional sonosensitizers doesn't trigger ferroptosis, the insight is about nanoparticle design, not SDT as a modality — and the regulatory path becomes device+drug+nanoparticle, the most complex category.

The ICD-to-immunity chain has repeatedly underperformed clinically. Pexa-Vec (oncolytic virus ICD, failed Phase 3) [M et al. [2019]], CheckMate-498 (immunotherapy in GBM, failed) [A et al. [2023]], KEYNOTE-361 (ICI+chemo in urothelial, failed) [T et al. [2021]], and BIND-014 (nanoparticle, failed Phase 2) [Von Hoff et al. [2018]] all demonstrate that preclinical rationale frequently fails in translation. SDT has no reason to be exempt from this base rate.

Narrow applicability. The hypothesis applies only to: accessible solid tumors (ultrasound can't treat diffuse metastatic disease), immunocompetent patients (ICD requires functional immunity), tumors that develop persister populations with OXPHOS/ferroptosis phenotype, and a specific post-therapy time window. Hematologic malignancies, brain tumors behind intact skull, and widely disseminated disease are excluded.

Evolutionary limitation. This hypothesis addresses one resistance transition. After SDT kills ferroptosis-sensitive persisters, survivors are selected for ferroptosis resistance (GPX4 upregulation, NRF2 activation, FSP1 restoration). The approach adds one step to the therapeutic arms race but does not end it. Durable eradication would require the ICD-triggered immune response to eliminate residual heterogeneous disease — the standard hope of all immunotherapy combinations, not specific to this hypothesis.

Prior art. Collateral sensitivity (Pluchino et al., 2012), adaptive therapy (Gatenby & Brown, 2009), and synthetic lethality are established frameworks for exploiting resistance tradeoffs. What our analysis adds is the specific SDT connection, not the general principle.

4.6 Research Priorities

Priority 1 (highest value, ∼3 months): Comparative ferroptosis killing experiment. Using persister cells from established models [M et al. [2026]], compare: SDT (with nanosonosensitizer) vs SDT (conventional sonosensitizer) vs RSL3 vs erastin. Measure cell death AND ICD markers (calreticulin, HMGB1, STING). This single experiment simultaneously tests (a) whether SDT kills persisters, (b) whether it produces superior ICD vs pharmacologic inducers, and (c) whether the nanosonosensitizer is required. Three questions, one experiment.

Priority 2 (∼1 month): Measure the ferroptosis vulnerability window. After drug

withdrawal from persister-inducing therapy, measure GPX4, FSP1, and ferroptosis sensitivity at 24h, 72h, 1 week, 2 weeks, and 4 weeks. If the window closes within days, the clinical feasibility of SDT intervention is severely limited. No published data quantifies this.

Priority 3 (if Priorities 1-2 positive, ∼6 months): SDT + anti-PD-1 vs RSL3 + anti-PD-1 in a syngeneic mouse model with OXPHOS-resistant tumors. Also include PDT + anti-PD-1 arm to directly test whether SDT outperforms the closest existing precedent. If SDT+ICI does not outperform PDT+ICI, the hypothesis of SDT's superiority collapses.

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5 Conclusion

The drug-tolerant persister ferroptosis field is searching for clinical ferroptosis inducers using only pharmacologic agents. This analysis of 10,413 articles proposes that physical ROS-generating modalities (PDT, SDT) should be evaluated as spatially selective alternatives that additionally produce immunogenic cell death.

The evidence base is substantial but disconnected: PDT has 355 PubMed ferroptosis articles, SDT has 121, and 44 papers document persister-cell ferroptosis sensitivity — yet these literatures barely cite each other. The connection at the modality-class level has not been systematically framed. Physical modalities offer spatial selectivity (focused energy at the tumor) and ICD generation (potential immunotherapy synergy) that systemic drugs do not. SDT specifically extends the approach to deep tumors that PDT cannot reach.

The decisive experiment: PDT vs SDT vs RSL3 on established persister cell models, measuring cell death AND ICD markers. If physical modalities kill persisters with equivalent efficacy and superior ICD, a new class of persister-targeting tools is validated. If they offer no advantage, the idea is dead.

We present this as a cross-literature perspective connecting two research communities that do not cite each other — not as a discovery, not as a breakthrough, and not as a cure. It is a research direction worth one experiment to test.

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